Ch 4 · Hypovolaemic shock and massive transfusion

Damage-control resuscitation

- Activate MTP; RBC:FFP:platelets ≈ 1:1:1 (PROPPR)
- Permissive hypotension (SBP 80–90) except TBI; minimise crystalloid
- TXA within 3 h (CRASH-2); pelvic binder, surgery/IR, REBOA in selected centres

Lethal triad (+ calcium)

- Warm fluids and patient; restore perfusion; fibrinogen > 1.5–2 g/L; ionised Ca > 1.1

Monitoring and complications

- TEG/ROTEM goal-directed therapy
- Hypocalcaemia, hyperkalaemia, hypothermia, TACO, TRALI

Key point: Early plasma and haemorrhage control beat large crystalloid volumes.

Ch 76 · Thrombotic thrombocytopenic purpura

Thrombotic microangiopathy differentials

- TTP (ADAMTS13 < 10%), STEC-HUS, atypical HUS, DIC, HELLP, malignant hypertension, drugs

Treatment

- Urgent plasma exchange (PLASMIC score; don't wait for ADAMTS13)
- Steroids + rituximab; caplacizumab (HERCULES)
- Avoid platelet transfusion unless life-threatening bleeding

Monitoring

- Platelets and LDH daily; troponin and neurological status; ADAMTS13 for relapse

Key point: Normal clotting with schistocytes and low platelets = TMA, not DIC.

Ch 77 · Acute promyelocytic leukaemia and DIC

Emergency

- t(15;17) PML-RARA; DIC with hyperfibrinolysis → early haemorrhagic death

Treatment

- ATRA immediately on suspicion + arsenic trioxide or chemotherapy
- Keep platelets > 30–50, fibrinogen > 1.5 g/L; avoid invasive procedures

Differentiation syndrome

- Fever, dyspnoea, infiltrates, effusions, weight gain, AKI
- Dexamethasone 10 mg 12-hourly; prophylaxis if WBC > 10

Key point: Give ATRA before genetic confirmation.

Ch 78 · Disseminated intravascular coagulation

Mechanism

- Tissue-factor-driven thrombin generation → microthrombosis + consumption → bleeding

Diagnosis

- ISTH score ≥ 5 (platelets, D-dimer, PT, fibrinogen); SIC score; serial tests

Treatment

- Treat cause
- Bleeding: platelets < 50, FFP for long PT, fibrinogen if < 1.5 g/L
- Not bleeding: platelets only if < 10–20; heparin for thrombosis-predominant DIC

Key point: Products treat bleeding, not numbers – the cause must be fixed.

Ch 79 · HLH and sepsis-like syndromes

Concept

- Uncontrolled macrophage/T-cell activation; triggers: EBV, lymphoma, autoimmune (MAS)

Diagnosis

- HLH-2004 ≥ 5/8; ferritin often > 10 000; sCD25; HScore > 169; marrow biopsy

Treatment

- Treat trigger; dexamethasone + etoposide (HLH-94)
- Anakinra for MAS/sepsis-like cases; IVIG; supportive care

Key point: Culture-negative 'sepsis' with very high ferritin and cytopenias – think HLH.

Ch 80 · HSCT complications

Problems

- Neutropenic sepsis, CMV, fungi; SOS/VOD; engraftment syndrome, DAH, IPS; GVHD, TMA

SOS/VOD

- Bilirubin ≥ 34, painful hepatomegaly, weight gain, ascites; reversed portal flow
- Defibrotide for severe; fluid and salt restriction, RRT

Acute GVHD

- Skin, liver, gut grading; methylprednisolone 1–2 mg/kg; ruxolitinib if refractory

Key point: Number of failing organs predicts ICU outcome after HSCT.

Ch 81 · Tumour lysis syndrome

Definition

- Cairo–Bishop: ≥ 2 of high urate, K, PO4 or low Ca; clinical = plus AKI, arrhythmia, seizure

Treatment

- Hydration 3 L/m²/day, UO > 100 mL/h
- Rasburicase (not in G6PD deficiency); allopurinol for prevention
- Treat high K; treat low Ca only if symptomatic; early RRT

No urinary alkalinisation

- Promotes calcium phosphate precipitation; not needed with rasburicase

Key point: Prevention starts before chemotherapy in high-risk tumours.

Ch 82 · Hyperviscosity syndromes

Causes

- Waldenström IgM, myeloma (IgA, IgG3), hyperleukocytosis, polycythaemia, cryoglobulins

Features

- Mucosal bleeding, visual change ('sausage' veins), neurological symptoms, heart failure

Treatment

- Plasmapheresis for paraprotein; leukapheresis + cytoreduction for leukostasis
- Avoid red cell transfusion until viscosity reduced; treat underlying disease

Key point: Transfusing red cells can worsen hyperviscosity.

Ch 83 · Thrombocytopenia in ICU

Causes

- Sepsis/DIC, dilution, drugs, HIT, ECMO/CRRT, TMA, liver disease, pseudothrombocytopenia

HIT

- 4Ts score; PF4/heparin immunoassay; functional assay confirms

HIT treatment

- Stop all heparin; argatroban, bivalirudin, fondaparinux or DOAC
- No warfarin until platelets > 150; no platelet transfusion; ≥ 3 months if thrombosis

Key point: HIT is prothrombotic – stopping heparin alone is not enough.

Ch 84 · Coagulopathy in liver disease

Rebalanced haemostasis

- Low pro- and anticoagulant factors; high vWF; INR does not predict bleeding

Assessment

- Viscoelastic tests; fibrinogen and platelets more useful than INR

Procedures

- No routine FFP for INR; ultrasound guidance, experienced operator
- High-risk procedures/bleeding: platelets > 50, fibrinogen > 1.2–1.5 g/L

Key point: Cirrhotic patients clot as well as bleed – still give VTE prophylaxis.

Ch 86 · Blunt abdominal trauma

Assessment

- ATLS; FAST positive + unstable → theatre; stable → contrast CT
- Seatbelt sign: bowel, mesentery, Chance fracture

Solid organ injury

- Non-operative management if stable; contrast blush → angioembolisation
- Serial exams; laparotomy if unstable/peritonitis; post-splenectomy vaccines

Damage-control surgery

- Control bleeding and contamination, temporary closure, ICU resuscitation, return in 24–48 h

Key point: Haemodynamic status, not injury grade, decides operation.

Ch 95 · Haemorrhagic shock

ATLS classes

- I < 15%; II 15–30% (tachycardia); III 30–40% (hypotension, confusion); IV > 40%

Resuscitation

- Pressure/tourniquet; early blood (O-neg → MTP 1:1:1), TXA, permissive hypotension
- Prevent hypothermia and hypocalcaemia; viscoelastic testing

Prehospital evidence

- PAMPer: prehospital plasma ↓ mortality; RePHILL: no benefit; whole blood evolving

Key point: Haemorrhage control is the definitive resuscitation.